

MARS AI

AI Employees on WhatsApp

Master Project Document

Product · Architecture · Packaging · Go-to-Market · Roadmap

Prepared for the MARS AI team

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Status: Living document — reflects build state as of June 2026

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1. Executive Summary

MARS AI is a multi-tenant platform that lets a small business “hire” AI employees that live entirely on WhatsApp. The first and flagship employee is Abdul, an AI accountant: the owner or staff send him expenses, income, receipts, voice notes, and udhaar (informal credit) over WhatsApp, and he keeps the company’s books, answers questions, generates documents, and produces reports — with no app to install, no forms to learn, and no English required.

The product is deliberately positioned not as “accounting software” but as “an AI employee with a salary.” This reframes the buying decision: instead of competing with ₹99/month apps, MARS AI is compared to a ₹15,000/month human accountant, against which a few thousand rupees a month reads as a bargain. Each additional agent (Abdul the accountant, Orion for correspondence, and future hires) is a new “employee” the customer can hire — clean, understandable expansion revenue.

The platform has been rebuilt from an early no-code (n8n) prototype into a real hosted application: TypeScript on Fastify, Supabase (PostgreSQL + Storage) for data, Google Gemini for the AI, and the WhatsApp Cloud API as the channel. It runs live on a production WhatsApp number today.

What this document covers

- Vision, positioning, and the market opportunity.
- The full product: every AI employee and every feature, in plain language.
- Packaging, pricing, the credit economics, and the free-trial design.
- Which services each package includes, and who is onboarded self-serve vs. by hand.
- The technical architecture, data model, and security/compliance posture.
- The customer dashboard and the internal admin console.
- The roadmap, current build status, risks, and an appendix reference.

IMPORTANT *This is a living internal document. Pricing figures marked “placeholder” are pending a forex reconciliation against a real Google invoice and have not been finalised for sale.*

Where the product stands today (June 2026)

- **Live & verified:**

Text/voice/photo logging, media handling, receipts, bulk statement import, document generation (invoices, quotations, letters on letterhead), Tally/GST export building blocks, inventory, reminders, reports, the Orion correspondence agent, the admin console, and a customer dashboard are all built and running. WhatsApp is on a real production number.

- **Recently built (SaaS foundation):**

Real per-user authentication with magic links, self-serve free-trial onboarding, a trial-aware customer dashboard with an upgrade path, a security/compliance audit log, and the khata (udhaar) free-tier hook.

- **Gated on business steps:**

Razorpay payments (needs company registration / KYC), lawyer-reviewed legal documents, and database row-level security (needs an architecture decision). These are the next focus.

2. Vision, Mission & Positioning

2.1 Vision

Every small business in India — and eventually every emerging market — should be able to hire a competent “back office” for the price of a phone recharge, operated entirely through the messaging app they already use all day. No software to learn, no language barrier, no laptop required.

2.2 Mission

Build a team of specialised AI employees that handle the operational, financial, and administrative work of a small business over WhatsApp — accurately, in the owner’s own language, and at a fraction of the cost of human staff or enterprise software.

2.3 Positioning — “AI Employees with Salaries”

MARS AI is marketed as hiring an AI employee and paying its salary, not as a software subscription. This is the single most important strategic decision in the go-to-market, adopted by the owner in May 2026. It does not change the underlying economics; it relabels the same machine with a far stronger story.

Why it wins: it changes the comparison anchor. As “software,” MARS AI competes with Vyapar at ₹99 and loses on price. As an “employee,” it competes with a ₹15,000/month human accountant — so ₹3,000/month reads as a steal. A non-technical owner instantly understands “hire a smart accountant for ₹3,000 instead of ₹15,000.”

The metaphor maps one-to-one onto the mechanics

Underlying mechanic	“Employee” label
Monthly subscription	Salary
Tier (Intern / Pro / Max)	Junior bookkeeper → Senior accountant → CA-grade
Included credit bucket	Fair-use workload
Overage credits	“Overtime” — your employee worked extra this month
Hiring Abdul + Orion together	Building a team / department (bundle discount)
Model tiers (Flash-Lite → Flash → Pro)	Why a senior costs more — he’s smarter

Three risks the metaphor introduces (and how we manage them)

1. “Salary” implies unlimited, but our cost is usage-based. A human caps at 8 hours a day; an AI does not. We never sell a flat salary with truly unlimited usage — we keep a fair-use workload plus “overtime” top-ups (the metaphor makes overage easy to sell).
2. “Employee” raises the accuracy bar to human level. App bugs are forgiven; a paid “employee” filing wrong GST gets fired loudly. This is fine for Intern/Pro, but the top “Max” tier’s sensitive features must be hardened before being sold as a CA-grade employee.

3. Multiple salaries can feel like nickel-and-diming, so we offer “hire the team” bundles that combine agents at a discount.

POSITIONING *The ₹15k-human anchor only lands on buyers who would actually employ an accountant (the small-growing-business beachhead). For solo owners it is pitched as “your first accountant,” priced like an intern.*

3. The Problem & Market Opportunity

3.1 The problem we solve

Tens of millions of Indian small businesses sit in a painful gap. They have outgrown the paper diary (the bahi-khata), but they are too small, too busy, or too non-technical to adopt Tally or even a forms-first app like Vyapar. Hiring a human accountant costs ₹10,000–20,000 a month and is overkill for their volume. So their books are messy, GST is a scramble, and they genuinely do not know who owes them money.

Why existing options fail this customer

- Paper diary / bahi-khata: zero learning curve, but nothing is searchable, no reports, no GST, no reminders, easily lost.
- Tally: powerful and the accountant's standard, but desktop-bound, complex, and requires accounting literacy the owner does not have.
- App-first SMB tools (Vyapar etc.): cheaper, but still demand that a busy owner open an app and fill structured forms in English — friction most never overcome.
- Khatabook / OkCredit: brilliant at the one udhaar feature, but they stop there — they are not a full back office and have no AI.

3.2 The wedge: udhaar (khata)

Indian shops run on udhaar — informal customer credit. Khatabook and OkCredit each reached tens of millions of users on this single feature: “who owes me, who do I owe, remind them.” It is a daily, vernacular, genuinely-needed behaviour — unlike “cleaner books,” which is merely nice to have. MARS AI uses khata as the free hook that builds a daily messaging habit, then expands into the deeper paid features as the business grows.

3.3 Why now

- WhatsApp is the universal interface in India — every shopkeeper already lives in it.
- LLMs are finally good and cheap enough to read receipts, transcribe code-switched Hindi-English voice notes, and reason about messy real-world inputs.
- GST has made “some kind of books” mandatory, expanding the pool of businesses that must track finances.

3.4 Defensibility

- A persistent, multi-tenant financial ledger with running balances and reminders is far more than a generic chat assistant — Meta's own WhatsApp Business AI will not run a company's books.
- The deep features (Tally/GST export, document generation on the customer's own letterhead, correspondence) are hard to build and exactly what an established firm pays for.
- Land-and-expand: khata gets them in free; as they grow they need staff, GST, and Tally — the higher tiers.

4. Beachhead & Go-to-Market

4.1 Who we sell to first

Segment	Pays enough?	Needs it enough?	Verdict
Personal / solo	No (₹100–250 ceiling)	No (diary survives)	Funnel only
Corner shop / cafe (1 person)	Barely	Nice-to-have	Funnel / land-and-expand
Small GROWING business	Yes (₹1,500–3,000/mo)	Yes — genuinely	★ BEACHHEAD
Enterprise / established firm	Highest	Yes, but errors non-negotiable	Premium upsell later

The beachhead is the small growing business: 2–10 staff, or 2–3 outlets, or GST-registered, or a vernacular owner who has outgrown the diary. This customer is too big for paper, too small or non-technical for Tally, and underserved by app-first tools. They both pay for the depth we built and genuinely need it.

4.2 Marketing by tier

- **Free / Intern — organic & viral:**

Vernacular short-form video (“text 50 chai → watch it logged”), with khata (“who owes you, on WhatsApp”) as the shareable hook. Goal is funnel, not revenue.

- **Shop tier — local & community:**

Market and trade WhatsApp groups, regional-language demos, “your accountant on WhatsApp — no app, no English.” Land-and-expand from free users.

- **Business (beachhead) — consultative & partner-led:**

Chartered Accountant referral partnerships (clean books mean less work for the CA), local business associations, case studies from the first customers, live demo calls.

- **Enterprise — sales-led:**

Founder-led outreach, pilots and proofs-of-concept, references built from the Business tier. This is where Tally/Orion depth (and later voice / an orchestrator agent) become the differentiators.

4.3 Break-even

Fixed platform cost is roughly ₹5,000–5,500 per month (hosting, database, WhatsApp number, domain). Break-even is approximately 16 Shop customers, or about 3–4 Business customers, or a single Enterprise customer. The first milestone is landing enough beachhead (Business) accounts to cover fixed cost — 3–4 paying Business customers makes the platform self-sustaining.

5. Product Overview — The AI Employees

MARS AI is a team of specialised agents. A company “hires” the agents it needs; each agent has its own system prompt, capabilities, and billing. The platform is multi-tenant — every company’s data is isolated, and an agent only ever sees the data of the company it is serving.

5.1 Abdul — the AI Accountant (flagship, live)

Abdul is the first and primary employee. Over WhatsApp he:

- Logs expenses and income from plain text (“paid 500 for diesel”, “received 20k from Aptitude”).
- Reads receipts and bills from photos and PDFs and logs them automatically.
- Transcribes voice notes (including code-switched Hindi-English) and acts on them.
- Imports whole bank/card statements (multi-page PDFs) and stages the transactions for review.
- Tracks khata / udhaar — informal running balances of who owes whom.
- Generates invoices, quotations, and bills on the company’s own template / letterhead.
- Handles salary and advances, employee approvals, and reversals (the ledger is append-only).
- Answers questions about the books and produces reports (Excel) on demand.
- Exports to Tally / GST formats for the company’s accountant (higher tiers).
- Tracks light inventory (stock quantities) and warns on low stock.
- Sets reminders that fire back to the owner on schedule.

5.2 Orion — the Correspondence Agent (live)

Orion handles written correspondence: emails, letters, and proposals on the company’s letterhead. Abdul hands off to Orion automatically when a request is about correspondence (intent-based routing, not name-matching). Orion drafts, the owner approves, and — once a Gmail connection is set up — it can send. It is offered as a paid add-on (“hire another employee”).

5.3 Nova — the Outreach Orchestrator (planned, parked)

Nova is a designed-but-deliberately-deferred “chief of staff” agent that would proactively message employees and clients, run goal-directed missions, assign tasks, and run a daily rhythm (morning to-dos, nightly brief). It is unlocked by the move to a paid WhatsApp number but is intentionally parked until the first product has paying customers — it is a third product before the first is monetised. It remains on the roadmap as a future Business/Enterprise upsell.

ARCHITECTURE *Design principle: one agent, many companies, strict per-company isolation. Adding a new capability is a new “marker” + handler; adding a new agent is a new hire the customer can make.*

6. Feature Catalog

This section describes every shipped capability in plain language. Each maps to one or more internal “actions” the agent can take.

6.1 Core logging

- **Text logging:**

Natural-language expense and income capture, auto-categorised, with entry IDs (EXP-, INC-).

- **Receipt / bill OCR:**

Photos and PDFs of receipts are read and logged; the file is saved to storage and indexed.

- **Voice notes:**

Transcribed with the higher-accuracy model (a mis-heard amount becomes a wrong ledger row, so transcription quality is prioritised) and then acted on like text.

- **Append-only ledger:**

Entries are never edited or deleted — corrections are made by reversing and re-logging, which keeps a clean audit trail.

6.2 Bulk statement import

A whole bank or card statement (multi-page PDF) is processed in the background: pages are read in chunks, transactions are extracted and staged in a review batch, and the owner commits all, some, or none with a single message. Heavy work runs as a durable background job so the chat stays responsive.

6.3 Khata / udhaar (free-tier hook)

Informal running balances per party, created on the fly from chat — distinct from invoices and from Tally. The owner says “Ramesh took 500 on udhaar” or “Ramesh paid 200,” and Abdul keeps a running balance (positive means they owe you, negative means you owe them). It is deliberately off-ledger: it tracks who-owes-whom, not profit and loss, to avoid double-counting. The owner can ask “who owes me?” and see it on the dashboard. Debtor auto-reminders (messaging the debtor directly) are a planned upgrade gated on the paid WhatsApp number and approved message templates.

6.4 Document generation

Invoices, quotations, letters, proposals, and purchase orders generated on the company’s own template. Two template mechanisms coexist: image/PDF templates use a vision-derived coordinate map, and Excel templates use a cell-address map. Values are stamped onto the blank template deterministically at render time (no AI at render time, so no hallucinated numbers). Line items are always treated as pre-tax and per-unit, with tax computed by the system.

6.5 Receivables (invoices)

A generated invoice becomes a receivable — a claim, not yet income. Income is logged only when the owner confirms payment, which closes the receivable and writes the income row in one step. This is separate from khata: receivables are formal invoices; khata is informal udhaar.

6.6 Salary, employees & approvals

Salary payments and advances are tracked with running advance balances and a linked expense entry. Staff can submit expenses for the boss to approve or reject, with WhatsApp notifications back to the requester. (These are Business-tier and above.)

6.7 Tally / GST export

The deepest feature: producing Tally-compatible vouchers and GST-ready data so the company's accountant can import clean books. This includes correct GST ledgers, inventory entries, and place-of-supply handling. It is the flagship Enterprise differentiator and a Business add-on.

6.8 Inventory

Lightweight stock tracking by quantity: the owner states opening stock and updates by message; sales logged through a bill deduct stock automatically; low-stock alerts surface on the dashboard and in chat. Batch/expiry tracking is deferred but on the roadmap.

6.9 Reminders & scheduled tasks

The owner can ask Abdul to remind them of anything; a background scheduler fires the reminder back over WhatsApp at the right time, supporting one-off and recurring schedules.

6.10 Reports

On-demand reports filtered by date / type, delivered as a formatted Excel workbook over WhatsApp (tinted header, currency formatting, totals). Custom Excel reports are kept free during the trial as a “wow” feature.

6.11 Correspondence (Orion)

Email/letter/proposal drafting on letterhead, with a draft → approve → send flow. Sending requires a Gmail connection per company.

6.12 Cross-cutting capabilities

- Reply-context: quoting an earlier WhatsApp message gives Abdul the context of that entry.
- Per-user preferred language: replies can be pinned to a language, or auto-match the user.
- Message batching: a rapid burst of messages (several photos + an instruction) is processed as one turn.
- Multi-agent routing: requests are routed to the right employee by intent.

7. Packaging & Pricing

IMPORTANT All rupee figures in this section are *PLACEHOLDERS* pending a forex reconciliation against a real Google invoice. They express the model and relative positioning, not final sale prices.

7.1 The economic reality

- 1 credit = ₹0.01 (one paisa).
- Credits are currently priced at Google’s list price converted at ₹83 per US dollar — i.e. the customer pays, in rupees, almost exactly what Google charges us. Gross margin on raw AI usage is therefore close to 0%.
- Implication: the subscription must be the profit; credits are a near-cost metered pass-through. Heavy users are funded by selling overage credits above cost.

Real cost per million tokens (verified model prices)

Model	Workload	Input ₹/1M	Output ₹/1M
gemini-2.5-flash-lite	OCR, statement extraction, bulk	8.3	33.2
gemini-2.5-flash	chat reasoning + voice transcription	24.9	207.5
gemini-2.5-pro	template analysis (one-time, absorbed)	103.75	830

Typical cost per interaction

Interaction	Approx credits	Approx ₹
Text log	~20	₹0.20
Voice note (transcribe + chat)	~22	₹0.22
Receipt photo (OCR + chat)	~22	₹0.22
Bulk statement (multi-page PDF)	~40–50	₹0.40–0.50

7.2 The tiers (“employees”)

Packaging is by persona (named tiers), not à-la-carte feature toggles. A customer cannot buy “inventory but not Tally” — that is a pricing and support nightmare. Features are bundled into tiers, with the credit meter layered on top (subscription base + usage overage).

Tier / “Employee”	Salary/mo (placeholder)	Included credits	Posture
Trial (free)	Free — 14 days or ~3,000 cr	3,000 cap	Funnel / hook
Abdul Intern (Lite)	₹499–999	15,000	Solo / shop
Abdul Pro ★	₹2,999	40,000	BEACHHEAD

Abdul Max	₹6,000–9,000 (custom)	100,000+	Enterprise
Orion (add-on)	₹1,999	30,000	Correspondence
“Hire the team” (Pro+Orion)	~₹3,999 bundle	combined	Team

MODEL POLICY *Model placement does NOT vary by tier. Every tier — the trial included — uses the same workload-based policy from §7.1: chat and voice on Flash, image/PDF OCR and statement extraction on the cheaper Flash-Lite, one-time template analysis on Pro. Tiers differ by features and included credits, never by which model runs a given task.*

Annual billing incentive: pay for 10 months, get 12 (two months free).

7.3 Where the margin comes from

The subscription is the profit; credits are a metered pass-through. Each tier includes a credit bucket sized so a typical user stays inside it — which also caps our downside (worst case we spend the whole bucket). Beyond the bucket, users buy overage credit packs at roughly three times our cost, so heavy users add profit instead of eroding it. After fair-use, blended margins on the paid tiers are in the 60–88% range.

7.4 Free-trial design

- Box: 14 days or about 3,000 credits, whichever comes first.
- No card required to start — lowest friction; the credit cap already caps our cost (about ₹30 per signup maximum).
- Same engine as paid — the trial runs the identical workload-based model policy, NOT everything on Flash. Chat and voice use Flash; image/PDF OCR and statement extraction use the cheaper Flash-Lite. A heavy document is therefore read on Flash-Lite, not Flash — so a single big file cannot torch the balance the way all-Flash would. What bounds trial cost is the credit cap, not a downgraded model. (Bulk statement import itself is paywalled — see below — so the largest multi-page jobs are not even available on the trial.)
- Free in the trial: full text/voice/photo logging, khata, custom Excel reports on demand, and the dashboard.
- Paywalled: bulk statement import, document generation, Tally/GST export, and multiple users.
- When the box is reached: a soft, three-stage landing (a warm heads-up near the limit, then a pause on billable actions with an upgrade prompt, then a read-only dashboard). Data is never deleted — it is the conversion magnet.
- One trial per WhatsApp number.

7.5 Three sign-offs still open before pricing is locked

4. Confirm the exact price points.
5. Validate the per-tier volume estimates against real usage data.
6. Reconcile the ₹83/USD assumption against one real Google invoice (decides whether we mark credits up or simply raise the conversion rate).

8. Services per Package

8.1 Capability matrix

Capability	Trial/Lite	Shop	Business ★	Enterprise
Text expense/income logging	✓	✓	✓	✓
Receipt / bill photo OCR	✓	✓	✓	✓
Voice notes	✓	✓	✓	✓
Khata / receivables (owner-facing)	✓ (hook)	✓	✓	✓
Monthly reports	basic / Excel	✓	✓ (xlsx)	✓ (xlsx)
GST-aware categorisation	—	✓	✓	✓
Bulk statement import	—	limited	✓	✓
Lite inventory (quantity)	—	✓	✓	✓
Multi-user + roles	—	2 users	✓	✓
Salary / employees / approvals	—	—	✓	✓
Document generation	— (free in trial)	—	✓	✓
Tally / GST export	—	—	add-on	✓
Orion correspondence	—	—	add-on	✓
Custom templates + priority SLA	—	—	—	✓

8.2 The onboarding split — a key decision

Self-serve onboarding is the free trial ONLY, and it collects the bare minimum. Every paid tier is onboarded manually by us on contact — precisely because the paid features (invoices, Tally, document generation, salary, correspondence) are what consume the heavy data (templates, client lists, products, vendors, GSTIN, letterheads). That data is collected by hand during a guided setup call, never in a self-serve form.

RULE *Rule of thumb: if a feature reads a master table or a template, that tier is onboarded manually; if it only writes the ledger (logging, khata), it is self-serve.*

Conversion is data-free: moving from the trial to the cheapest paid tier needs no new data — the customer simply adds payment. Only the deeper features trigger a manual setup.

9. Onboarding & Customer Lifecycle

9.1 Self-serve trial signup (built)

A public landing page leads to a six-field signup form: business name, WhatsApp number, and consent are required; name, business type, and preferred language are optional. No card, no email, no GSTIN, no templates. On submit, the system provisions a trial company (with a credit grant and a 14-day expiry), creates the owner as the boss user, seeds sensible expense/income categories based on the business type, and records the consent (for data-protection compliance).

9.2 Activation (built)

The success page shows a “message us on WhatsApp” deep link that opens WhatsApp with a pre-filled “hi”. Because the user messages first, the welcome — a friendly greeting plus a tap-to-login dashboard link — lands inside WhatsApp’s 24-hour service window, so no special approved template is needed. The user’s first real message (“50 chai”) is still processed, so their first log is not swallowed. Activation is defined as the first successful log.

9.3 Login — magic links (built)

There are no passwords. The customer dashboard is opened via a single-use, time-limited magic link delivered over WhatsApp (the user messages “login,” or an operator sends one). Tapping it creates a revocable session. This is both more secure and friendlier than sending credentials over chat.

9.4 The trial → paid journey

- In-trial nudges: as the trial is used up (around 70%), the owner gets a warm “enjoying Abdul? reply upgrade” message; the dashboard shows days left and a credit-usage bar.
- Upgrade: because paid plans are set up by hand, “upgrade” routes to a guided contact rather than a self-serve checkout (for now).
- Expiry: an expired trial is paused (not deleted) with an upgrade prompt; the data is preserved as the conversion magnet.

9.5 Manual onboarding (paid tiers)

For paid customers, we run a short setup: collect GSTIN/PAN and address, upload letterhead and invoice/quotation templates, import the client and product/vendor lists, and connect Tally or Gmail if relevant. This is high-touch by design and is exactly the work the paid features need.

10. Technical Architecture

10.1 Technology stack

Layer	Choice
Language / runtime	TypeScript on Node.js
Web framework	Fastify
Database	Supabase PostgreSQL
File storage	Supabase Storage
AI / LLM	Google Gemini (2.5 Flash, Flash-Lite, Pro)
Messaging channel	WhatsApp Cloud API (Meta)
Hosting	Render (web + background worker)
Email (Orion)	Gmail API via OAuth
Documents	pdf-lib + ExcelJS; LibreOffice for conversion

10.2 Request pipeline (how a message is handled)

7. A WhatsApp webhook delivers an inbound message to the web service.
8. Rapid bursts are debounced into a single “turn” so multiple attachments plus an instruction are seen together.
9. The sender is identified and mapped to a company and user; unknown numbers get one polite “not registered” reply.
10. Service/credit gates run (paused account, zero credits, trial expiry, login keyword, first-contact welcome).
11. Media is downloaded, stored, and analysed: images/PDFs are OCR-read, voice notes transcribed.
12. Context blocks are assembled (company profile, account info, open batches, receivables, khata balances, Tally ledger names, reminders, reply-context, history).
13. The right agent is resolved (honouring explicit mentions / intent routing) and the LLM is called.
14. The agent’s reply is parsed for “markers” (structured action requests); safety nets catch cases where the model claims an action without emitting its marker.
15. Each action runs through its handler (writing the ledger, recording khata, generating a document, etc.), and the reply — plus any attachments/notifications — is sent back over WhatsApp.
16. Usage is billed in credits; messages, errors, and audit events are recorded.

10.3 Agents, markers & actions

Each agent has a system prompt describing its behaviour and the “markers” it may emit — short structured tags like [LOG_EXPENSE] or [LOG_KHATA] embedded in its reply. A shared parser turns markers into typed actions, and a registry maps each action to a handler. Adding a capability is therefore: define a marker, write a handler, add a prompt clause, and add a safety net. The safety

net is essential because the model will occasionally claim it did something without emitting the marker; the net detects the claim and surfaces the failure rather than silently losing data.

10.4 Model policy (cost control)

Different workloads use different Gemini models on purpose. Chat reasoning runs on Flash; voice transcription also uses Flash (accuracy matters); image/PDF OCR and statement extraction use the cheaper Flash-Lite; one-time template analysis uses Pro. These are deliberately decoupled so that improving chat quality does not silently make every receipt scan four times more expensive.

10.5 Background work

Heavy or long-running tasks (bulk statement extraction, scheduled reminders, the Tally export queue, trial-expiry sweeps) run on a background worker on a periodic tick, using safe locking so a doubled runner cannot double-fire. A per-user processing lock prevents concurrency races on a single conversation.

10.6 Multi-tenancy

Every record carries a `company_id`, and every query is scoped to the authenticated user's own company — a company can never see another's data. Database-enforced row-level security is the next security hardening step (see §11).

11. Security, Privacy & Compliance

11.1 Data protection (India DPDP Act)

India's Digital Personal Data Protection Act applies from the very first customer, with no size threshold. MARS AI is a Data Fiduciary with five core duties: obtain explicit consent before collecting data; delete within seven days of a verified request; notify the Data Protection Board of a breach within 72 hours; name a grievance contact; and publish a privacy policy. Consent is now captured at signup (policy version, timestamp). General personal data may be stored abroad (Supabase), but payment data must stay in India — so we never store card data ourselves and will use a payment processor (Razorpay) that holds it in India.

11.2 WhatsApp channel policy

The product rides on a channel we do not own, so policy compliance is existential. We never ask users for full card, bank-account, or government-ID numbers in chat. Proactive outbound messages (khata reminders, future orchestration) require approved utility templates and respect the 24-hour window. Wherever possible, flows are designed so the user messages first (e.g. the signup activation deep link and the login keyword), which keeps replies inside the service window with no template dependency.

11.3 Authentication & sessions (built)

- Customer dashboard: passwordless, single-use, 15-minute magic links over WhatsApp; tapping one mints a revocable 30-day session. Tokens are stored only as hashes, so a database leak cannot be replayed into a login.
- Admin console: a separate operator authentication, isolated from customer access.
- Access can be revoked per user (kills all sessions).

11.4 Audit log (built)

A dedicated audit trail records security and compliance events — logins, billing/service changes, data exports, and signups — separate from error logs and usage logs. It is viewable in the admin console and supports data-protection accountability.

11.5 Secrets & isolation

- Per-company credentials (e.g. Tally, Gmail) are stored in an encrypted vault.
- All traffic is over HTTPS; the database encrypts data at rest.
- Errors are captured to a persistent store and surfaced in the admin console.

11.6 The remaining security items

- **Row-Level Security (RLS):**

Database-enforced tenant isolation so a missing filter in code cannot leak one company's data to another. This needs an architecture decision because the app currently connects with a privileged key that bypasses RLS; enabling true enforcement means changing how the app authenticates to the database. This is the next security build.

- **Legal documents:**

Privacy Policy, Terms of Service, and a Data Processing Agreement — currently simple stubs, to be replaced with lawyer-reviewed copy.

- **Payments:**

Razorpay integration (never storing cards), gated on company registration / KYC.

- **Later:**

SOC 2 only when an enterprise customer demands it; a formal data-retention/purge policy for the audit log.

12. Admin Console & Customer Dashboard

12.1 Three surfaces

Surface	Who	Notes
Marketing site	Public	Landing + “Start free”; fuller site (mission/products/docs) later
Super-admin console	Us only	Master control of all companies; separate auth
Customer dashboard	Customers	One role-aware app; boss-first

Decision: a web app (responsive, installable as a PWA) plus WhatsApp — no native app. The product is WhatsApp; the dashboard is occasional-use (reports, billing, settings), for which the web is ideal and avoids app-store friction.

12.2 Super-admin console (largely built)

Operator control over every company: overview, per-agent message and token/credit stats, message and error logs, the new audit log, editable system prompts, onboarding instructions, and a masked view of environment configuration. Per company there are tabs for summary, agents, documents, templates, clients, vendors, products, employees, outbound approvals, credentials, data sources, Tally, ledger, receivables, batches, files, workspace jobs, messages, errors, and settings (edit profile, categories, top up credits, pause/resume, manage users, send/revoke login links).

12.3 Customer dashboard (built, role-aware)

One dashboard that renders by role rather than three separate apps:

Role	Sees
Boss / Owner	Everything: full ledger, reports, billing/credits, manage AI employees & users, settings
Manager	Operational: ledger, reports, approvals, team activity — no billing, no user management
Employee	Minimal: only what they submitted and their own usage

Roles matter only at the Pro/Max tiers (a solo customer is one person), so the dashboard is boss-first at launch, with manager and employee views layered in later. Today the dashboard includes an overview (with a trial banner showing days left and credit usage), the ledger with an Excel download, khata, receivables, inventory, reminders, documents, files, the workspace (job progress), messages, and an upgrade page.

13. Data Model (overview)

The schema is managed by forward-only numbered migrations. The principal tables:

Table	Purpose
companies	Tenant root: profile, plan, trial flags, credit balances, settings
users	WhatsApp users of a company (boss/employee), preferred language, sessions
agents / company_agents	Agent definitions and which company has hired which agent
ledger_entries	The append-only expense/income/salary ledger
receivables	Invoices issued and awaiting payment
khata_parties / khata_entries	Informal udhaar running balances and their history
clients / vendors / products	Master data for paid features (invoicing, inventory)
employees	Read-only HR master (salary, joining)
files	Index of all stored files (receipts, generated docs, voice notes)
messages	Every inbound/outbound message per company
sessions	Per-user agent conversation state
batches / pending_transactions	Bulk statement staging and review
scheduled_tasks / jobs	Reminders and durable background jobs
credit_ledger / token_usage_log	Credit movements and per-call token/billing records
company_documents / _template_maps	Uploaded assets and analysed template coordinate/cell maps
company_credentials / _data_sources	Encrypted secrets vault and linked data sources
magic_links / portal_sessions	Passwordless auth (single-use links, revocable sessions)
audit_log	Security/compliance events
consents	Data-protection consent records captured at signup
app_errors	Persistent error log surfaced in admin
tally_* tables	Tally integration: ledger sync, invoices, outbox

NOTE Balance-sign convention for khata: a positive party balance means they owe you; negative means you owe them.

14. Roadmap & Build Status

14.1 SaaS foundation scorecard

Item	Status
Per-user auth + magic links	Built · deployed · live-verified
Self-serve trial onboarding	Built · deployed
Trial-aware customer dashboard + upgrade page	Built · deployed
Security/compliance audit log	Built · deployed
Khata / udhaar (free-tier hook)	Built · deployed
Legal documents (Privacy/ToS/DPA)	Stubs live; lawyer copy pending
Razorpay payments	Gated on company registration / KYC
Row-Level Security (RLS)	Pending an architecture decision

14.2 Near-term build order

17. Database row-level security (tenant isolation enforced at the database).
18. Razorpay payments in test mode now, ready to flip live once registration lands.
19. Lawyer-reviewed legal documents.
20. Debtor-facing khata reminders (once the paid number + approved templates are ready).
21. Manager/employee dashboard views as the first Pro customers land.

14.3 Deferred / later

- Nova, the outreach orchestrator agent (parked until the first product is monetised).
- Native voice calls (enterprise-only, later).
- Inventory batch/expiry tracking.
- Salary monthly accrual; the remaining advanced salary/budget/trust controls.
- A fuller marketing site (mission, products, docs, changelog).

14.4 Open business decisions

22. Confirm final price points.
23. Validate per-tier volume estimates against real usage data.
24. Reconcile the forex assumption against a real Google invoice.
25. Decide who drafts the legal documents (template service vs. lawyer).
26. Register the operating entity (Private Limited / LLP) and GST.

15. Risks & Mitigations

Risk	Mitigation
Zero AI margin if priced at cost	Subscription is the profit; credits are near-cost; overage sold at ~3× cost
Possible 2× undercharge via forex (₹83/USD)	Reconcile one month of usage against a real Google invoice before locking pricing
WhatsApp policy violation → channel restriction/ban	User-first flows, approved templates for proactive sends, never request card/ID numbers in chat
Data-protection (DPDP) non-compliance	Consent at signup, 7-day deletion, never store cards (Razorpay), audit log, legal docs
AI errors at the “CA-grade” tier	Harden sensitive features before selling the Max “employee”; keep humans-in-the-loop on approvals
Cross-tenant data leak	Strict company-scoped queries today; database row-level security next
Model cost spikes on heavy media	Model-tiering (cheap model for OCR/extraction), credit caps, background jobs
Single-channel dependency (WhatsApp)	Channel-agnostic message model internally; web dashboard as a second surface
Free-tier abuse	One trial per WhatsApp number; credit + time box; soft block, never delete data

16. Appendices

Appendix A — Glossary

Term	Meaning
Agent / AI employee	A specialised AI persona (Abdul, Orion) a company can hire.
Marker	A structured tag in an agent's reply (e.g. [LOG_EXPENSE]) that triggers an action.
Action / handler	The code that executes a marker (writes the ledger, records khata, etc.).
Credit	The billing unit; 1 credit = ₹0.01, sized to roughly one unit of Google AI cost.
Khata / udhaar	Informal customer credit; a running balance of who owes whom.
Receivable	A formal issued invoice awaiting payment (distinct from khata).
Ledger	The append-only record of expenses, income, and salary.
Magic link	A single-use, time-limited passwordless login link.
Tier	A persona-based package (Trial, Intern, Pro, Max) plus the Orion add-on.
Beachhead	The first target customer segment: small growing businesses.
Data Fiduciary	The DPDP term for an entity that determines how personal data is processed.

Appendix B — Agent action reference (selected markers)

Marker	What it does
LOG_EXPENSE / LOG_INCOME	Write an expense / income ledger entry
LOG_KHATA	Record an udhaar entry and update a party balance
LOG_SALARY	Record a salary/advance with running balance + linked expense
REVERSE_EXPENSE / REVERSE_INCOME / REVERSE_SALARY	Reverse a prior entry (append-only correction)
MARK_INVOICE_PAID	Close a receivable and log the income on payment
GENERATE_DOCUMENT / GENERATE_REPORT	Produce a document or an Excel report
LOG_SALES_BILL	Extract and log a photographed sales bill
UPDATE_STOCK	Adjust inventory quantities
BULK / COMMIT_BULK	Stage and commit bulk-statement transactions
APPROVE_EXPENSE / REJECT_EXPENSE	Boss decision on a submitted expense
SCHEDULE_TASK / COMPLETE_TASK / CANCEL_TASK	Reminders and scheduled tasks
TALLY_REPORT	Read live figures from the company's Tally
ROUTE_TO	Hand off to another agent (e.g. Orion for correspondence)

Appendix C — Document history

This document is generated from the project's own planning records: the packaging & pricing strategy, the SaaS foundation plan, the self-onboarding plan, the khata plan, the rebuild plan, and the running build log. It reflects the build state as of June 2026 and should be regenerated as the product evolves.

— *End of document* —

17. Product in Action — Sample Conversations

The following illustrative dialogues show how the product feels in everyday use. They are representative examples, lightly idealised, to help the team picture the experience. All happen inside WhatsApp.

17.1 Logging an expense by text

Everyday capture

Owner: paid 1200 for diesel cash
Abdul: Logged EXP-0042 — ₹1,200 diesel, cash. 📄
Owner: got 25000 from Sharma Traders by upi
Abdul: Logged INC-0031 — ₹25,000 from Sharma Traders, UPI.

17.2 A receipt photo

Owner: [sends a photo of a restaurant bill]
Abdul: Read your bill — ₹2,360 at Cafe Coffee Day, 02-Jun-2026. Logged EXP-0043 under Meals. Want me to tag it to a project?

17.3 A voice note (Hindi-English)

Owner: [voice note] "bhai aaj 500 ka chai paani hua, cash"
Abdul: Logged EXP-0044 — ₹500 chai-paani, cash. 🎧

17.4 Khata / udhaar

The free-tier hook

Owner: Ramesh took 500 on udhaar
Abdul: Noted. 📄 Ramesh now owes you ₹500.
Owner: Ramesh ne 200 diye
Abdul: Got it. 📄 Ramesh now owes you ₹300.
Owner: who owes me money?
Abdul: Here's your khata:
• Ramesh — owes you ₹300
• Anil — owes you ₹1,500
You owe: • Suresh — ₹1,000

17.5 Generating an invoice

Owner: make an invoice for Aptitude, website development, 50000 plus gst
Abdul: Here's your invoice on your letterhead. 📄 INV-0007 — Aptitude Pvt Ltd, Website Development, ₹50,000 + 18% GST = ₹59,000. I've added it to receivables; tell me when they pay and I'll book the income.
System: [PDF invoice delivered as a WhatsApp document]

17.6 Confirming payment of an invoice

Owner: Aptitude paid the website invoice

Abdul: Marked INV-0007 paid (₹59,000) and logged it as income INC-0032. Receivable closed. 

17.7 A bulk bank statement

Background processing

Owner: [sends a 9-page bank statement PDF]

Abdul: Reading your statement — this one's big, so I'll work in the background and ping you. 

Abdul: Done — I found 84 transactions in BATCH-3. Reply "commit BATCH-3" to log them all, or open the dashboard to review first.

Owner: commit BATCH-3

Abdul: Logged all 84 transactions (₹3.4L expenses, ₹5.1L income). 

17.8 Asking a question / report

Owner: how much did we spend on fuel this month?

Abdul: ₹14,700 on fuel in June so far, across 11 entries.

Owner: send me this month's expenses as excel

Abdul: Here's your June expense report. 

System: [formatted .xlsx workbook delivered]

17.9 A reminder

Owner: remind me to pay GST on the 18th

Abdul: Reminder set (TASK-12): "Pay GST" on 18-Jun, 10:00 IST. I'll message you then.

17.10 Handoff to Orion (correspondence)

Intent-based routing

Owner: draft a thank-you email to Aptitude for the project

Abdul: Passing this to Orion — correspondence is his desk, not mine.

Orion: Here's a draft thank-you email to Aptitude. Want me to adjust the tone, or send it as is once your Gmail is connected?

17.11 Trial nudges and upgrade

Abdul:  You've used most of your free trial (820 of 3,000 credits left). Enjoying Abdul? Reply "upgrade" and we'll keep him working for you — your data stays put.

Owner: login

Abdul:  Your secure login link: <https://.../portal/login?token=...> (works once, expires in 15 min).

18. Feature Deep-Dives

This section expands the feature catalogue with the design intent, the flow, and the important edge cases for each major capability.

18.1 Logging & the append-only ledger

Every financial fact lands in a single ledger of expense, income, and salary rows, each with a stable entry code (EXP-, INC-, SAL-). The ledger is append-only: nothing is ever edited or deleted. Corrections are made by reversing the original entry (which writes a contra row) and logging the correct version. This guarantees a defensible audit trail — critical for a financial product — and avoids the class of bugs where an “update” silently loses history.

Edge cases handled

- Ambiguous amounts or missing payment mode: Abdul asks a short clarifying question rather than guessing.
- Duplicate-looking entries: reply-context lets the owner point at a specific prior entry.
- A claimed log without an actual write: a safety net detects the model claiming “Logged EXP-...” without the underlying action and surfaces the failure instead of losing the transaction.

18.2 Media: photos, PDFs, and voice

Inbound media is downloaded, stored in per-company storage, indexed in a files registry, and analysed. Images and PDFs are read by the vision model; voice notes are transcribed by the higher-accuracy model because a mis-heard amount becomes a wrong ledger row. The Meta media CDN occasionally returns transient errors, so downloads retry several times. A burst of several files plus an instruction is debounced into one turn so Abdul sees everything together.

18.3 Bulk statement import

A multi-page statement is too large for a single model call and too slow to block the chat, so it runs as a durable background job: pages are chunked, transactions extracted in parallel, and the results staged into a review batch. The owner commits all, a subset, or cancels. Progress is visible on the dashboard workspace. Because it is a background job, a worker restart resumes rather than restarts the work.

18.4 Khata / udhaar (design detail)

Khata is the free-tier hook and is deliberately distinct from both invoices and Tally. It tracks an informal running balance per party, created on the fly by name (no client master needed). The balance is signed: positive means the party owes the business, negative means the business owes the party. Four intents map onto the balance: credit given (they owe more), payment received (they owe less), borrowed (you owe more), and payment made (you owe less). It is off-ledger by design — it never writes income or expense, so it cannot double-count profit and loss. The owner can ask “who owes me?” at any time, and a dashboard tab lists both sides with totals. Debtor-facing reminders (messaging the debtor) are a planned upgrade gated on the paid number and approved templates; “remind me to chase Ramesh” already works today.

18.5 Document generation

The owner's real invoice/quotation/letter template is uploaded once and analysed into a map: for image/PDF templates, a vision model produces normalised coordinates of where each value goes; for Excel templates, a map of cell addresses. At runtime the agent emits the values, and a deterministic renderer stamps them onto the blank template — no AI at render time, so no hallucinated numbers or layouts. Line items are always treated as pre-tax and per-unit; the system computes tax, so a quotation reused as an invoice never double-applies GST. Fonts (including the rupee glyph) are embedded so output is faithful across devices.

18.6 Tally / GST export

The deepest and most defensible feature. The system can produce Tally-compatible sales vouchers and GST-ready data: correct GST ledgers (CGST/SGST for intra-state, IGST for inter-state, by place of supply), inventory entries with stock item, quantity, unit, and rate, and HSN details — the things that make a return acceptable. It reads live figures from a connected Tally for questions like “what's our cash balance?”, “who owes us?”, “show the P&L”. Final GST sign-off requires a licensed Tally (the educational mode blocks some operations), which is acceptable since the accountant imports into their own licensed copy.

18.7 Reports

Reports are produced as formatted Excel workbooks (tinted frozen header, auto-filter, currency formatting, a totals row, greyed reversed rows) and delivered over WhatsApp. They can be filtered by date and type. Custom Excel on demand is kept free even in the trial as an impressive “wow” feature.

18.8 Reminders & scheduling

Reminders are stored and fired by a background scheduler on a one-minute tick, supporting one-off and recurring schedules (daily, weekly, weekdays) in IST. They currently fire to the owner; third-party reminders await the paid-number capability.

19. Unit Economics — Worked Examples

IMPORTANT Illustrative, using placeholder prices and the current ₹83/USD assumption. Final numbers await the forex reconciliation.

19.1 A typical Pro customer's month

Assume an Abdul Pro customer (₹2,999/month, 40,000 included credits) running a moderately active business at roughly 1,000 interactions in the month.

Line	Amount
Subscription revenue	₹2,999
Typical usage (~32,000 credits)	≈ ₹320 of Google cost
Gross profit if inside the bucket	≈ ₹2,679 (~89%)
Worst case (whole 40,000-credit bucket spent)	≈ ₹400 cost → ₹2,599 profit (~87%)

19.2 Overage (“overtime”) example

If the same customer has a heavy month and exceeds the bucket by 20,000 credits, those are sold as an overage pack at roughly three times cost (about ₹0.03/credit ≈ ₹600), of which roughly two thirds (≈ ₹400) is margin. Heavy users therefore add profit rather than eroding it.

19.3 Trial cost

A trial is capped at about 3,000 credits ≈ ₹30 of Google cost — the maximum we can spend acquiring one trial signup, since the credit cap is also the cost cap. No card is required because the cap already bounds the downside.

19.4 Platform fixed cost & break-even

Item	≈ ₹/month
Hosting (web + worker)	~2,100
Database	~2,100
WhatsApp number / messaging	~500–1,000
Domain / misc	~300
Total fixed	~5,000–5,500

Break-even is roughly 3–4 paying Business (Pro) customers. Past that, each additional Pro customer is almost pure margin, because the fixed cost is already covered and per-customer AI cost is small.

20. Go-to-Market Playbook

20.1 The funnel

27. Awareness: vernacular short-form video and WhatsApp-group demos showing a message becoming a logged entry, and khata answering “who owes me.”
28. Trial: one-tap “Start free” → minimal form → WhatsApp activation. No card.
29. Activation: the first successful log (instrumented as the key early metric).
30. Habit: daily khata + logging use builds the messaging habit everything rides on.
31. Conversion: in-trial nudges + the dashboard upgrade path; paid tiers onboarded by hand.
32. Expansion: add Orion (“hire another employee”), move up tiers as the business grows, add Tally/GST at the top.

20.2 Channels by segment

- Solo/shop: organic, viral, community — funnel, not revenue.
- Business beachhead: CA referral partnerships, business associations, case studies, demo calls.
- Enterprise: founder-led sales, pilots, references from Business customers.

20.3 The CA partnership angle

Chartered Accountants are a force multiplier: clean books from their clients mean less work for the CA, and the Tally/GST export hands them data in the format they already use. A CA who onboards several clients is both a distribution channel and a trust signal.

20.4 What NOT to do

- Don’t build Nova (the orchestrator) before the first product has paying customers.
- Don’t offer à-la-carte feature toggles — persona tiers only.
- Don’t lock prices before the forex reconciliation.
- Don’t chase enterprise logos before the accuracy bar is proven on the Business tier.

21. KPIs & Metrics to Track

Metric	Why it matters
Trial signups	Top of funnel; effectiveness of awareness channels
Activation rate (first log)	Predicts retention; the key early indicator
Daily active (messages/day)	Habit strength — especially khata usage
Trial → paid conversion	The core business metric
Credits per customer per month	Validates bucket sizing and unit economics
Gross margin per tier	Confirms the subscription-is-profit model holds
Overage revenue	Whether heavy users self-fund
Churn / pause rate	Product-market-fit signal
Error rate per 1,000 turns	Quality bar for the “employee” promise
Support contacts per customer	Operational load and onboarding quality
Forex-adjusted cost/credit	Guards against silent margin loss

PRIORITY *Instrument activation (first successful log) and trial→paid conversion first — those two numbers decide everything else.*

22. Operations & Support Runbook

22.1 Deployment

The application auto-deploys from the main branch to the host. Forward-only database migrations apply on deploy. The agent system prompt lives in the database, so prompt changes require a seed step on deploy to take effect. After deploy, the standard verification pass is: confirm the health endpoint, run a signup, and click through the dashboard and audit log.

22.2 Monitoring & debugging

- The admin console's Messages tab shows every inbound/outbound message per company.
- The Errors tab surfaces captured exceptions (also written to a persistent store).
- The Audit tab shows security/compliance events (logins, billing changes, exports, signups).
- Hosting logs provide the real-time view for live debugging.

22.3 Common operational tasks

Task	How
Onboard a paid company	Admin "Add company", then the setup tabs (templates, clients, etc.)
Top up / adjust credits	Company → Settings → top up (recorded in the credit ledger + audit)
Pause / resume a company	Company → Settings → service status
Send a customer their dashboard link	Company → Users → "Send login link"
Revoke a user's access	Company → Users → "Revoke access" (kills all sessions)
Edit an agent's system prompt	Admin → Agents → edit prompt
Investigate a "lost" message	Messages tab + Errors tab + hosting logs

22.4 Known operational notes

- On the free hosting tier the service can sleep after idle and cold-start; the first message after a deploy can land during a restart window. A paid tier removes this.
- Proactive (cold) outbound needs approved WhatsApp templates; user-first flows avoid this.
- Prompt changes are not live until the seed step runs on deploy.

23. Build History (Condensed Timeline)

A condensed history of how the product reached its current state.

Period	Milestone
Early prototype	No-code (n8n) version of Abdul on WhatsApp: text loop, receipts, salary, approvals, bulk statements, credit accounting
Rebuild begins	Re-platform to TypeScript + Fastify + Supabase + Gemini (the “real app”)
Phases 0–2	Text loop, media (images/PDFs/voice), files registry, send-files-back — live
Phase 3	Reversals, salary, approvals, bulk extract/commit, reports
Admin portal	Multi-tab operator console; per-company management; editable prompts
Phase 6	Document generation via analysed templates (PDF/image + Excel)
Tally integration	GST ledgers, sales vouchers, inventory entries, live reports
Inventory	Quantity-based stock tracking with low-stock alerts
Production WhatsApp number	Switched from test to a real production number
SaaS foundation (current)	Magic-link auth, self-serve trial onboarding, trial-aware dashboard + upgrade, audit log, and khata — all built

NOTE *The early no-code workflows are retained only as a behavioural reference; the live product is the rebuilt application.*

24. Extended Appendices

Appendix D — Competitive positioning

Option	Strength	Why MARS AI wins
Paper diary	Zero learning curve	Searchable, reports, GST, reminders, never lost
Tally	Accountant standard	No accounting literacy needed; lives in WhatsApp
Vyapar / app-first	Cheap, structured	No app, no forms, no English; conversational + AI
Khatabook / OkCredit	Great at udhaar	Full back office + AI, not just one feature
Generic WhatsApp AI	Convenient chat	Persistent books, ledger, documents, Tally — real work
Human accountant	Trusted, flexible	A fraction of the cost; always-on; instant

Appendix E — Marker / action payloads (selected)

Marker	Key fields
LOG_EXPENSE	date, description, vendor, category, amount, payment mode, reference
LOG_INCOME	date, description, source, category, amount, payment mode, reference
LOG_KHATA	party, kind (credit_given / payment_received / borrowed / payment_made), amount, note, date
LOG_SALARY	employee, type (salary / advance / repayment), amount, date
MARK_INVOICE_PAID	invoice_code, payment_mode, date
GENERATE_DOCUMENT	kind, values (per the analysed template's fields)
GENERATE_REPORT	type, date range, filters
UPDATE_STOCK	product, change / set-to quantity, unit
SCHEDULE_TASK	title, schedule (once / recurring), time
ROUTE_TO	agent (e.g. orion), reason

Appendix F — Schema migration history (selected)

Migration	Adds
0001 init	Core tables: companies, users, agents, ledger, sessions, messages
0002–0005	Error log; company assets; onboarding wizard; template maps
0006–0013	Clients; CIN/HSN; products; employee columns; approvals; Gmail drafts; vendors
0014–0020	Receivables; reversal contras; chat-model upgrade; Tally integration &

	invoices
0021–0027	Message buffer; scheduled tasks; portal token; processing lease; jobs; inventory; language
0028 magic auth	Magic links + revocable portal sessions
0029 trial & consent	Trial flags/expiry + consent records
0030 audit log	Security/compliance audit trail
0031 khata	Khata parties + entries (udhaar balances)

Appendix G — Frequently asked questions

Q. Is there an app to download?

A. No. The product is WhatsApp. The dashboard is a web app (optionally installable as a PWA) for occasional use — reports, billing, settings.

Q. What languages does it support?

A. It understands and replies in the owner’s language, including code-switched Hindi-English; a preferred language can be pinned per user.

Q. How is my data isolated from other businesses?

A. Every record is tagged to your company and every query is scoped to it; database-level row security is the next hardening step.

Q. Do you store my card?

A. No. Payments are handled by a processor that holds card data in India; we never store it.

Q. What happens when my trial ends?

A. Service pauses with an upgrade prompt — your data is preserved, never deleted.

Q. Can my staff use it too?

A. Yes, on paid tiers, with roles (manager/employee) and approvals.

Q. Can my accountant get the data into Tally?

A. Yes — Tally/GST export is the top-tier feature.

— continued —

25. Full Feature Catalogue

Every capability the platform ships today, grouped by area. This is the canonical list the team should use when describing what the product does.

34.1 Capture & logging

Feature	Description
Text logging	Log expense/income from a plain message in natural, code-switched language.
Receipt photos	Read a photo of a bill and log it with amount, vendor, date, category.
PDF documents	Read PDF bills/invoices/statements.
Voice notes	Transcribe voice (higher-accuracy model) and log the transaction.
Multi-file bursts	Several files plus an instruction are reasoned over together.
Reply context	Quote a previous message to point the agent at a specific entry.
Append-only ledger	Stable codes (EXP-/INC-/SAL-); corrections by reversal + relog.

34.2 Money management

Feature	Description
Reversals	Reverse expense/income/salary; writes a contra row, preserves history.
Salary & advances	Log salary, advances, and repayments with a running advance balance.
Approvals	Boss approves/rejects employee-submitted expenses; requester is notified.
Receivables / invoices	Issue an invoice, track it as a receivable, mark it paid → income.
Khata / udhaar	Informal running balances per party; who owes whom; off-ledger.
Bulk statement import	Background extraction of many transactions; review & commit.
Inventory	Quantity-based stock with low-stock alerts; deduction on sale.

34.3 Documents & reporting

Feature	Description
Document generation	Invoices/quotations/letters on the owner's own analysed template (no AI at render).
Excel reports	Formatted, filterable .xlsx reports delivered over WhatsApp.
Tally / GST export	Tally-compatible sales vouchers, GST ledgers, inventory & HSN; live Tally reads.
Send files back	Retrieve and resend any stored file on request.

34.4 Productivity & coordination

Feature	Description
Reminders & scheduling	One-off and recurring reminders fired by a background scheduler (IST).
Orion (correspondence)	Draft emails/letters/proposals; intent-based handoff from Abdul.
Multi-user & roles	Boss/manager/employee roles with approvals (paid tiers).
Preferred language	Per-user pinned language; otherwise matches the owner's style.

34.5 Platform, billing & access

Feature	Description
Credit billing	Token-based credits; buckets + overage; thinking tokens counted.
Magic-link auth	Texted one-time login → revocable dashboard session.
Customer dashboard	Web view of books, ledger, receivables, khata, inventory, reminders, settings.
Admin portal	Internal operator console across all companies and agents.
Audit log	Security/compliance event trail.
Self-serve trial	Public signup → minimal data → WhatsApp activation.

26. Personas & Tiers — Detail

The product is sold as AI employees you “hire”, each with a salary (a monthly price) and a workload (an included credit bucket). Customers pick the employee and the seniority, not a feature checklist.

35.1 The “employees with salaries” framing

- **Abdul:** Abdul — the Accountant. Books, documents, reports, GST.
- **Orion:** Orion — the Correspondent. Emails, letters, proposals.
- **Nova:** Nova — the Chief of Staff (future). Outreach, missions, coordination.

Seniority maps to tiers: an Intern does the basics, a Pro does the full job, a Max handles the heaviest and most specialised work (e.g. Tally/GST). Hiring a second employee is an upsell, not a plan change.

35.2 Abdul — tier feature matrix (illustrative)

Capability	Intern	Pro	Max
Text / photo / voice logging	✓	✓	✓
Append-only ledger & reversals	✓	✓	✓
Khata / udhaar	✓	✓	✓
Excel reports	✓	✓	✓
Reminders	✓	✓	✓
Multi-user & approvals	—	✓	✓
Document generation (templates)	—	✓	✓
Receivables / invoicing	—	✓	✓
Inventory	—	✓	✓
Bulk statement import	—	✓	✓
Tally / GST export	—	—	✓
Priority / higher bucket	—	—	✓

NOTE The matrix is the intended shape of the tiers; exact inclusions and prices are finalised with the forex reconciliation before launch. Persona tiers only — no à-la-carte feature toggles.

35.3 Pricing model (structure, not final numbers)

Element	How it works
Monthly salary	A fixed price per employee per tier — the subscription.
Included bucket	Monthly credits sized so typical usage stays well inside, keeping margin high.
Overage packs	Usage beyond the bucket, sold at roughly 3× cost so heavy users self-fund.

Second employee	Add Orion (or another) for an additional salary — the main expansion motion.
Trial	Free, ~3,000 credits, minimal data, pauses on lapse — a funnel, not a tier.

35.4 Segment → tier guidance

Segment	Likely tier	Why
Solo / small shop	Trial → Intern	Capture, khata, simple reports; price-sensitive
Growing business (beachhead)	Pro	Documents, invoicing, multi-user, inventory — the sweet spot
Established / GST-heavy / CA-served	Max	Tally/GST export, highest volume, priority

27. Admin Portal Walkthrough (Internal)

The admin portal is the operator console used by the MARS AI team to run the platform. Access is by a login cookie. It has a platform-wide section and per-company tabs.

36.1 Platform-wide

View	Purpose
Overview	Health and headline numbers across all companies.
Agents	All agents; edit each agent's system prompt.
Messages	Every inbound/outbound message across companies (debugging).
Errors	Captured exceptions from the pipeline and worker.
Audit	Security/compliance events (logins, exports, billing, signups, access changes).
Meta-config	Masked view of environment configuration.
Instructions	Editable onboarding and add-agent guides.
Companies list	Entry point to per-company management; new-company form.

36.2 Per-company tabs

Tab	Purpose
Summary	The company at a glance.
Agents	Per-agent message counts, tokens, credits; assign/unassign agents.
Documents	Upload logos, letterheads, profile, signatures.
Templates	Paired blank+example template uploads; per-kind Analyze → placement map.
Credentials	Encrypted secret vault (e.g. Tally/Gmail).
Data sources	Link or upload Excel/Sheet/DB pointers.
Ledger	The company's financial entries.
Files	Stored media/documents registry.
Messages	This company's message history.
Errors	This company's captured errors.
Settings	Edit company/categories; top up credits; pause/resume; manage users + login links.

28. Customer Dashboard Walkthrough

The dashboard is the customer-facing web view. It is optional for daily use — the product is WhatsApp — but valuable for reports, review, billing, and settings. Access is via a texted magic link.

Tab	What the customer sees / does
Overview	Headline stats; on a trial, a banner with days left, a credit-usage bar, and Upgrade.
Ledger	All entries; download a formatted .xlsx export.
Receivables	Outstanding invoices and their status.
Khata	Two lists — they owe you / you owe them — with totals.
Inventory	Current stock levels (read-only).
Reminders	Scheduled reminders.
Documents	Generated and uploaded documents.
Messages	Their conversation history.
Upgrade	What upgrading unlocks; a guided contact link (paid = manual onboarding).
Settings	Profile, categories, users, language, billing/top-up.

GROWTH NOTE *The dashboard is the funnel's exit for trials: the overview banner and the upgrade page convert a happy trial into a guided paid onboarding.*

29. Onboarding Playbooks

38.1 Self-serve trial (automated)

33. Prospect taps “Start free” on the landing page → the public start form.
34. They enter business name, WhatsApp number, and consent (optional: contact name, business type, language). No card, email, GSTIN, or templates.
35. The system provisions a trial company (trial plan, ~3,000 credits, boss user, consent record), and seeds category presets for the chosen business type.
36. The success page shows a wa.me link; the prospect messages the number to activate.
37. First inbound → welcome + magic link, then the first real log proceeds.
38. In-trial nudges warn at ~70% credit usage (“reply upgrade”); lapse pauses service with an upgrade prompt, never deleting data.

38.2 Paid onboarding (manual, high-touch)

39. Prospect contacts the team (often via the dashboard upgrade link or a referral).
40. Team creates the company in the admin portal and sets the plan and credits.
41. Team loads the setup data via the company tabs: company profile and categories; documents (logo/letterhead); templates (upload blank+example, then Analyze); clients, products, vendors; credentials (Tally/Gmail) and data sources as needed.
42. Team adds the company’s users and sends each a login link.
43. A guided first session confirms logging, a generated document, and a report.
44. Handover: the customer now self-serves day-to-day on WhatsApp; the team monitors via Messages, Errors, and Audit.

WHY MANUAL *Paid tiers are onboarded by hand precisely because they need templates, masters, and integrations — the things that make the higher tiers valuable.*

38.3 CA / partner onboarding

45. Identify a CA serving multiple in-segment businesses.
46. Show the Tally/GST export — the data lands in the format they already use.
47. Onboard their clients as paid companies; the CA becomes both a channel and a trust signal.

30. Team Roles & Responsibilities

A lightweight responsibility map so everyone knows who owns what. Roles, not necessarily distinct people, at this stage.

Area	Owns	Supports
Product & strategy	Owner	Whole team
Engineering / build	Owner + AI pair	—
Prompt & agent quality	Owner	Engineering
Sales & partnerships (CA)	Owner	Future hire
Onboarding (paid)	Owner	Support
Support & operations	Owner	Future hire
Finance / pricing / forex	Owner	—
Compliance (DPDP, WhatsApp)	Owner	Legal (external)

NOTE *This document, the build log, and the decisions log exist specifically to reduce key-person dependency as the team grows.*

31. Legacy (No-Code) Reference

Before the rebuild, the product ran as two linked no-code (n8n) workflows. They are retained only as a behavioural reference — the live product is the rebuilt application — but the team should know the history.

Workflow	Role
Channel Router	Received the raw WhatsApp webhook, validated user/company, enriched the payload, and routed to the right AI employee.
Abdul Accountant	Received the payload, talked to the model, and wrote to per-company Google Sheets.

The no-code version proved the concept (text loop, receipts, salary, approvals, bulk statements, credit accounting) but could not be properly tested, secured, or scaled — which is why the product was re-platformed. Crucially, the new schema was designed around the old data’s problems, so the rebuild started fresh rather than importing the old books.

— End of operational reference —

32. Technical Architecture — The Request Lifecycle

This section traces a single WhatsApp message end-to-end so an engineer joining the team understands exactly how the system behaves, in order.

25.1 The pipeline, step by step

48. Webhook receipt: WhatsApp (Meta) posts the inbound message to the application's webhook. The request is acknowledged immediately so Meta does not retry; the real work is queued.
49. Identification: the sender's number is matched to a user and a company. Unknown senders get one throttled "not registered" reply. A trial boss's very first message triggers a welcome + login link.
50. Static gates (no AI, no billing): login/dashboard keyword → magic link; service-paused → notice; zero/low credits → role-aware message. These short-circuit before any model call so they cost nothing.
51. Debounce / batching: a burst of messages or files within a short window is coalesced into one turn so the agent reasons over the whole context, not fragments.
52. Media handling: any attached media is downloaded (with retries), stored, registered in the files table, and analysed (vision for images/PDFs, transcription for voice).
53. Context assembly: the pipeline builds the prompt context — company profile, account/credit info, reply context, language directive, and conditional blocks (open batches, pending receivables, khata balances, scheduled tasks, available templates, clients/products/vendors, Tally ledgers). Conditional blocks appear only when relevant, keeping token cost down.
54. Per-user lock: a processing lease prevents two concurrent turns for the same user from racing.
55. Agent run: the selected agent (Abdul by default) calls the model with its system prompt + context + history, and returns a reply that may contain action markers.
56. Marker parsing: the shared parser extracts markers (e.g. LOG_EXPENSE) and maps them to action handlers via the agent's declared marker map.
57. Safety nets: before acting, the agent checks for claims-without-markers (the model saying it did something it did not emit) and for cross-domain intents (correspondence → route to Orion).
58. Action execution: each handler runs against the database (append-only ledger writes, khata updates, document generation, report building, etc.), producing effects.
59. Billing: token usage (including thinking tokens) is converted to credits and deducted; the usage is logged. Mixed-model turns are billed at the higher rate by deliberate policy.
60. Reply + effects delivery: the natural-language reply and any generated files are sent back over WhatsApp. Errors are captured to the error store; security-relevant events to the audit log.

25.2 Why a worker and durable jobs

Light turns complete inline. Heavy turns (multi-page statements, large documents) are dispatched to a durable background job so the chat is never blocked and a process restart resumes rather than restarts. The job's steps and progress are visible in the dashboard workspace.

25.3 The billing engine

One credit equals one thousand tokens. The engine counts prompt, completion, and thinking tokens, maps them through the model's rate, and deducts credits atomically alongside a usage-log row.

Buckets (monthly included credits) are drawn down first; overage is sold in packs priced at roughly three times cost. Template-analysis calls are logged but not charged. This design makes a subscription almost pure margin while protecting against runaway cost.

25.4 The model policy

A single policy module decides which model handles which workload. Conversational turns and voice transcription use the higher-capability model (a mis-heard amount is expensive); images, PDFs, bulk extraction, and contact OCR use the cheaper, fast model. These are deliberately decoupled and must not be recombined, because the cost/accuracy trade-off differs per workload.

33. Data Dictionary

A plain-language description of the main data the system keeps. Exact column lists live in the schema migrations; this is the conceptual model the team should share.

26.1 Core entities

Entity	What it holds
Company	A tenant: name, type, plan, credit balance, service status, settings (categories, GSTIN, profile), trial flags and expiry.
User	A person on a company's WhatsApp: phone, name, role (boss/manager/employee), preferred language, portal session/token.
Agent	An AI employee (Abdul, Orion): key, display name, editable system prompt, model, enable/assignment per company.
Session	The rolling conversational state per user/agent: recent history, buffer.
Message	Every inbound/outbound message with direction, content, media link, tokens, and the agent that handled it.

26.2 Financial entities

Entity	What it holds
Ledger entry	The append-only financial record: code (EXP-/INC-/SAL-), date, description, party, category, amount, payment mode, reference, reversed flag, contra link.
Receivable	An issued invoice awaiting payment: invoice code, client, amount, status, line items.
Khata party	An informal counterparty with a signed running balance (+ they owe you / – you owe them).
Khata entry	An append-only udhaar movement: kind, amount, note, date, balance-after.
Token usage log	Per-turn billing record: tokens by type, model, credits charged.
Credit ledger	Top-ups and deductions to a company's credit balance.

26.3 Operational entities

Entity	What it holds
File	Registry of stored media/documents: type, storage path, linked message, company.
Batch / pending transaction	Staged bulk-extract results awaiting commit.
Scheduled task	A reminder: title, schedule (once/recurring), next fire time, status.
Job / job step	A durable background job and its steps, for heavy work and the workspace view.
Product / stock	Catalogue items and quantity-based stock levels with low-stock thresholds.

Client / vendor	Customer and supplier masters (paid tiers): name, aliases, code, GSTIN, contacts.
Employee	Staff records for salary/advance tracking (when seeded).
Template map	The analysed placement map (coordinates or cell addresses) per company template kind.

26.4 Security, compliance & onboarding entities

Entity	What it holds
Magic link	A 15-minute, single-use login token, stored only as a hash.
Portal session	A 30-day revocable dashboard session, stored only as a hash.
Consent	A DPDP consent record captured at signup (who, what, when).
Audit log	Security/compliance events: logins, exports, billing changes, signups, access changes.
Error log	Captured exceptions from the pipeline and worker for debugging.
Credential vault	Per-company secrets (e.g. Tally/Gmail), encrypted at rest.

ARCHITECTURE NOTE Multi-tenant isolation is enforced today by always scoping every query to a company id. Database-level row security (RLS) is the next hardening step and is gated on a deliberate decision about the service-role connection.

34. Security, Privacy & Compliance

27.1 Authentication

Dashboard access uses magic links: the owner texts “login”, receives a one-time link valid for fifteen minutes, and lands in a thirty-day revocable session. Tokens are never stored in plaintext — only their hashes — so a database leak cannot be replayed into a login. Access can be revoked instantly, killing all sessions for a user.

27.2 Secrets

Per-company credentials (such as Tally or email connections) live in an encrypted vault. A dedicated encryption key is configured in the environment so encryption is stable across restarts.

27.3 Data protection (DPDP)

Consent is captured at signup and stored as a record. Personal data is minimised: the trial collects only a business name, a WhatsApp number, and consent. Data is never deleted on trial lapse — service pauses instead — and an audit trail records security-relevant actions. Privacy and Terms pages exist as stubs and will be replaced with reviewed copy.

27.4 Tenant isolation

Every record carries a company id and every query is scoped to it. The roadmap’s row-level-security step will push isolation into the database itself as defence in depth.

27.5 Messaging compliance

Within the 24-hour service window the system can reply freely. Proactive (cold) outbound requires approved WhatsApp message templates; the product is designed so the common flows are user-initiated (e.g. activation via a wa.me link, login via a keyword), avoiding template friction.

27.6 Cost & abuse safeguards

- Hard credit caps bound the cost of any single account, including trials.
- A monthly cloud budget cap with hard-disable protects against runaway AI spend.
- Unknown senders and low/zero-credit states are handled before any paid model call.

35. Prompt Engineering Philosophy

The agents are reliable not because the model is perfect, but because the system is engineered to catch the model's failure modes. The team should internalise these principles before changing agent behaviour.

28.1 Markers, not free-form actions

Agents request actions by emitting structured markers (e.g. LOG_EXPENSE with a JSON payload). The application — not the model — performs the action. This keeps a clean boundary between “what the model wants” and “what actually happened.”

28.2 The safety-net pattern (mandatory)

HARD RULE *Every new marker requires three things together: a prompt clause teaching the agent when to emit it, an action handler, and a regex safety net that catches the model claiming the action without emitting the marker. Without the net, the model will occasionally lie about completing work and the data silently won't be written.*

When a claim-without-marker is detected, the agent overrides its own reply to tell the truth (“I said I'd do X but didn't actually record it”) rather than leave the owner believing a false success.

28.3 Context blocks, conditionally injected

Relevant state is injected as labelled blocks (pending receivables, khata, open batches, available templates...) only when it exists. This keeps each turn cheap and focused, and lets the agent answer state questions (“who owes me?”) directly and authoritatively instead of computing from raw history.

28.4 Intent-based routing, not name-matching

Handoff between agents is driven by intent (any correspondence → Orion), never by fuzzy matching of a spoken name, because voice transcription mangles names. A symmetric safety net on each side reinforces the routing.

28.5 Disambiguation rules

Where two features look similar (khata vs invoice vs income; salary vs advance), the prompt contains explicit “pick the right one” rules and instructs the agent to ask a short question rather than guess — guessing is what produced double-booked salary advances before the rule was added.

28.6 The prompt lives in the database

OPERATIONAL GOTCHA *Because the system prompt is stored in the database and read at runtime, prompt changes are not live until the deploy seed step runs. This is the single most common “why didn't my change take effect?” gotcha.*

36. Risk Register & Mitigations

Risk	Impact	Mitigation
AI cost spikes / forex drift	Margin erosion	Credit caps; monthly budget hard-disable; pending forex reconciliation before locking prices
Model hallucinates an action	Lost/wrong data, broken trust	Marker + safety-net pattern; append-only ledger; override reply on claim-without-marker
Wrong amount from voice/photo	Bad books	Higher-accuracy model for voice; clarifying questions; easy reversal
Tenant data leak	Severe trust/legal	Company-scoped queries now; RLS next; hashed tokens; encrypted secrets
Free-tier host cold starts	Dropped/slow first message	Move to paid host tier; retries; durable jobs resume
WhatsApp policy violation	Number ban	User-initiated flows; approved templates for cold outbound
Payments blocked (no Pvt Ltd/KYC)	Can't auto-bill	Manual payment links until incorporation/KYC
Over-scoping (build Nova too early)	Wasted runway	Strict roadmap order: prove paid customers first
Tally educational-mode wall	GST sign-off blocked	Customer/CA imports into their licensed Tally
Key-person dependency	Bus factor	This document; build log; decisions log; code comments

37. Detailed Roadmap

30.1 SaaS foundation (in progress)

Item	Status	Note
Per-user auth + magic link	Done, live	Revocable sessions; hashed tokens
Self-serve trial onboarding	Done, deploying	Public start form; minimal data; wa.me activation
Trial-aware dashboard + upgrade	Done	Banner, credit bar, upgrade page
Audit log	Done	Security/compliance events + admin view
Khata / udhaar	Done	Free-tier hook; owner-facing v1
Legal docs (Privacy/Terms)	Stubs live	Reviewed copy later
Row-level security (RLS)	Next	Needs service-role-connection decision
Payments (Razorpay)	Blocked	Needs Pvt Ltd + KYC; manual links meanwhile

30.2 Product polish (optional, opportunistic)

- Debtor-facing khata reminders (gated on paid number + templates).
- Native .xlsx everywhere; salary monthly accrual; release-salary / budget / trust handlers.
- Dashboard auth via the new sessions (replace any legacy token paths fully).
- Richer marketing site; reviewed legal copy.

30.3 Growth bets (after first paying customers)

61. Orion to full email send (connect Gmail) — correspondence becomes end-to-end.
62. Deeper inventory (batches/expiry) — owner wants this eventually.
63. Nova, the outreach orchestrator — chief-of-staff agent that messages staff/clients and runs goal-directed missions; unlocked by the production number; explicitly after product-market fit.
64. More AI employees (the “hire another employee” expansion motion).

STRATEGY Roadmap order is a strategic commitment, not a backlog. Foundation before growth; paying customers before Nova; forex reconciliation before locking prices.

38. Key Decisions & Rationale

Rebuild from no-code to a real app

The no-code prototype proved the concept but could not scale, be tested, or be hardened. The TypeScript application is the product; the old workflows are reference.

Start fresh on data (no migration)

The new schema was designed around the old data's problems; the old system was already paused, so importing it would import its flaws.

Persona tiers, not feature toggles

Customers buy an “employee with a salary”, not a feature menu. Tiers are simpler to sell, price, and reason about.

Self-serve = trial only

Paid tiers need real setup (templates, clients, products) and are onboarded by hand; only the minimal-data trial is self-serve.

Off-ledger khata

Udhaar is a balance, not profit/loss; keeping it off-ledger prevents double-counting and keeps the books clean.

Append-only ledger

Corrections via reverse-and-relog preserve a defensible audit trail.

Mixed-model overage billed high

A deliberate policy choice: heavy/mixed turns should fund themselves; this is intended, not a bug to fix.

Magic-link auth over permanent links

A permanent token in a URL is a standing liability; short-lived, hashed, revocable sessions are the right default.

Foundation before growth features

Identity, billing rails, isolation and compliance must exist before scaling acquisition or building Nova.

39. Glossary

- Abdul** — The accountant AI employee — the first and primary agent.
- Orion** — The correspondence AI employee — drafts and (soon) sends emails/letters.
- Nova** — The planned outreach orchestrator (“chief of staff”) agent — future, post-PMF.
- Agent** — An AI employee with its own persona, prompt, and capabilities.
- Marker** — A structured tag the agent emits to request an action (e.g. LOG_EXPENSE).
- Action handler** — The code that performs the work behind a marker.
- Safety net** — A check that catches the model claiming an action it did not emit.
- Context block** — A labelled chunk of state injected into the prompt only when relevant.
- Credit** — The billing unit: one credit = one thousand tokens.
- Bucket** — A tier’s monthly included credits.
- Overage / overtime** — Usage beyond the bucket, sold in packs at roughly three times cost.
- Khata / udhaar** — Informal running credit balances with named parties (the free-tier hook).
- Receivable** — An issued invoice awaiting payment (distinct from khata).
- Ledger** — The append-only record of expense, income, and salary entries.
- Reversal / contra** — How a ledger entry is corrected without deleting history.
- Bulk extract** — Background extraction of many transactions from a statement, staged for commit.
- Template map** — The analysed placement of values onto a document template.
- Magic link** — A one-time, short-lived, hashed login link.
- Session** — A revocable dashboard login (also the per-user conversational state).
- Trial** — A free, capped, minimal-data self-serve plan that pauses on lapse.
- DPDP** — India’s Digital Personal Data Protection framework — the basis for consent handling.
- RLS** — Row-level security — database-enforced tenant isolation (planned).
- Tally** — The accounting software Indian businesses/CAs use; the top-tier export target.
- GSTR-1 / HSN / GSTIN** — GST return, product classification code, and taxpayer id, respectively.
- Admin portal** — The internal operator console for managing companies and agents.
- Customer dashboard** — The customer-facing web view of their books, reports, and settings.

40. How to Use & Maintain This Document

This is a living document generated from the project's own planning and build records. To keep it trustworthy:

- Treat the build log and decisions log as the primary sources of truth; this document summarises them.
- Regenerate it from the generator script when the project reaches a milestone, rather than editing the Word file by hand (hand-edits are lost on regeneration).
- Update the page-number fields in Word (select all, press F9) after opening, to refresh the table of contents.
- Keep distribution internal — it contains strategy, economics, and roadmap.

Questions about any section should go to the owner (Umar Iqbal). Engineering specifics live alongside the code in the build log, decisions log, and the per-feature plan documents referenced throughout.

— *End of document* —